

COMPUTER NETWORKS LAB

Experiment 1: Networking Devices

1. Hub

A hub is a basic networking device that connects multiple computers. It broadcasts incoming data to all connected devices without checking the destination address.

[Paste a clear picture of Hub here]

2. Switch

A switch connects devices in a LAN and forwards data only to the correct destination using MAC addresses. It creates a dedicated path for communication, significantly improving network efficiency.

[Paste a clear picture of Switch here]

3. Router

A router connects different networks and directs data packets using IP addresses. It enables Internet access by intelligently determining the best path for data to travel.

[Paste a clear picture of Router here]

4. Bridge

A bridge connects two LAN segments and filters traffic using MAC addresses. It helps reduce broadcast traffic and improve network performance.

[Paste a clear picture of Bridge here]

5. Modem

A modem converts digital signals to analog signals and vice versa to connect to an ISP. It links a home or office network to the Internet.

[Paste a clear picture of Modem here]

6. Repeater

A repeater regenerates weak signals and retransmits them to extend network distance and maintain signal quality.

[Paste a clear picture of Repeater here]

7. Gateway

A gateway connects networks using different protocols and translates data between them.

[Paste a clear picture of Gateway here]

8. Network Interface Card (NIC)

A NIC enables a computer to connect to a wired or wireless network and serves as the hardware interface.

[Paste a clear picture of Network Interface Card (NIC) here]